

# Circular Economy Ontology Network (CEON) - Resource ODP

## Metadata

**IRI**

<http://w3id.org/CEON/ontology/resourceODP/>

**Title**

Circular Economy Ontology Network (CEON) - Resource ODP

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**License**

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**Version Iri**

<http://w3id.org/CEON/ontology/resourceODP/0.3/>

**Version Info**

0.3

**Preferred Namespace Prefix**

resourceODP

**Preferred Namespace Uri**

<http://w3id.org/CEON/ontology/resourceODP/>

**Description**

A core ODP of the CEON ontology network defining aspects of the resource concept.

**Covers Requirements**

Covers the following requirements from Onto-DESIDE D3.1: CVN-Resource-1,3, CVN-Composition-1,2, CVN-ResrouceType-4, C7-3, E2-2, E4-6, E5-1, E6-3, T3-1.

## Classes

|                       |                                                                                                               |
|-----------------------|---------------------------------------------------------------------------------------------------------------|
| Location <sup>C</sup> |                                                                                                               |
| IRI                   | <a href="http://w3id.org/CEON/ontology/location/Location">http://w3id.org/CEON/ontology/location/Location</a> |
| In Range Of           | <a href="#">hasResourceLocation</a> <sup>Op</sup>                                                             |

## Batch of objects<sup>C</sup>

|                     |                                                                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resourceODP/BatchOf0bjects">http://w3id.org/CE0N/ontology/resourceODP/BatchOf0bjects</a>                                                                                                                             |
| <b>Description</b>  | A batch of objects is a collection of physical objects that are of the same type, e.g. a set of items (product objects) adhering to the same product model.                                                                                                 |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup><br><a href="#">has physical object</a> <sup>op</sup> only <a href="#">Physical object</a> <sup>C</sup> <i>and</i> <a href="#">has physical object</a> <sup>op</sup> some <a href="#">Physical object</a> <sup>C</sup> |
| <b>In Domain Of</b> | <a href="#">batch size</a> <sup>dp</sup>                                                                                                                                                                                                                    |
| <b>In Range Of</b>  | <a href="#">has batch</a> <sup>op</sup>                                                                                                                                                                                                                     |
| <b>Restriction</b>  | <a href="#">batch size</a> <sup>dp</sup> <i>exactly</i> 1 <a href="#">Batch of objects</a> <sup>C</sup>                                                                                                                                                     |

## Constituent<sup>C</sup>

|                    |                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>         | <a href="http://w3id.org/CE0N/ontology/resourceODP/Constituent">http://w3id.org/CE0N/ontology/resourceODP/Constituent</a> |
| <b>Description</b> | A constituent is a component of object.                                                                                   |
| <b>In Range Of</b> | <a href="#">has constituent</a> <sup>op</sup>                                                                             |

## Digital object<sup>C</sup>

|                     |                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resourceODP/Digital0bject">http://w3id.org/CE0N/ontology/resourceODP/Digital0bject</a> |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup>                                                                                         |

## Information<sup>C</sup>

|                     |                                                                                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IRI</b>          | <a href="http://w3id.org/CE0N/ontology/resourceODP/Information">http://w3id.org/CE0N/ontology/resourceODP/Information</a>                                                                      |
| <b>Description</b>  | Information is an abstract concept that represents any kind of interpretations. For instance, information can be data generated by software systems or data used by people for communications. |
| <b>Sub Class Of</b> | <a href="#">Resource</a> <sup>C</sup>                                                                                                                                                          |
| <b>In Domain Of</b> | <a href="#">containsInformation</a> <sup>op</sup><br><a href="#">isAbout</a> <sup>op</sup>                                                                                                     |
| <b>In Range Of</b>  | <a href="#">containsInformation</a> <sup>op</sup><br><a href="#">is realization of</a> <sup>op</sup>                                                                                           |

## Matter<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/Matter>

**Description** A matter is a physical substance.

**In Range Of** [has matter](#)<sup>op</sup>

## Physical object<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/PhysicalObject>

**Description** A physical object is a collection of matter.

**Sub Class Of** [Resource](#)<sup>C</sup>  
[has constituent](#)<sup>op</sup> only [Constituent](#)<sup>C</sup> and [has constituent](#)<sup>op</sup> some [Constituent](#)<sup>C</sup>  
[has matter](#)<sup>op</sup> only [Matter](#)<sup>C</sup> and [has matter](#)<sup>op</sup> some [Matter](#)<sup>C</sup>

**In Domain Of** [has constituent](#)<sup>op</sup>  
[has matter](#)<sup>op</sup>

**In Range Of** [has physical object](#)<sup>op</sup>

## Resource<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/Resource>

**Description** A resource able to be handled in the context of a circular value network, e.g. data generated by software systems in the CVN, materials or products as physical objects handled in the CVN.

**In Domain Of** [hasResourceLocation](#)<sup>op</sup>

**Super Class Of** [Batch of objects](#)<sup>C</sup>  
[Digital object](#)<sup>C</sup>  
[Information](#)<sup>C</sup>  
[Physical object](#)<sup>C</sup>  
[Set of objects](#)<sup>C</sup>

## Set of objects<sup>C</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/SetOfObjects>

**Description** A set of objects is a set of physical objects (items) that can be of different types, i.e. different kinds of items.

**Sub Class Of** [Resource](#)<sup>C</sup>  
[has batch](#)<sup>op</sup> some [Batch of objects](#)<sup>C</sup> and [has batch](#)<sup>op</sup> only [Batch of objects](#)<sup>C</sup>  
[has physical object](#)<sup>op</sup> only [Physical object](#)<sup>C</sup> and [has physical object](#)<sup>op</sup> some [Physical object](#)<sup>C</sup>

**In Domain Of** [has batch](#)<sup>op</sup>

## Object Properties

has location<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/location/hasLocation>

**Super Property Of** [hasResourceLocation<sup>op</sup>](#)

contains information<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/containsInformation>

**Sub Property Of** [hasPart<sup>op</sup>](#)

**Domain** [Information<sup>c</sup>](#)

**Range** [Information<sup>c</sup>](#)

has batch<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/hasBatch>

**Description** hasBatch intends to represent that a set of objects can be captured by a number of batches where each batch contains a number of physical objects.

**Domain** [Set of objects<sup>c</sup>](#)

**Range** [Batch of objects<sup>c</sup>](#)

has constituent<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/hasConstituent>

**Description** hasConstituent intends to represent that a physical object can have a collection of composing components.

**Sub Property Of** [hasPart<sup>op</sup>](#)

**Domain** [Physical object<sup>c</sup>](#)

**Range** [Constituent<sup>c</sup>](#)

has matter<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resourceODP/hasMatter>

**Description** hasMatter intends to represent that a physical object can have a collection of matter.

**Domain** [Physical object<sup>c</sup>](#)

**Range** [Matter<sup>c</sup>](#)

## has part<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/hasPart>

**Super Property Of**

- [containsInformation](#)<sup>op</sup>
- [has constituent](#)<sup>op</sup>

## has physical object<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/hasPhysicalObject>

**Description**

hasPhysicalObject intends to represent that a batch of objects or a set of objects can have composing components of physical objects.

**Domain**

[Batch of objects](#)<sup>c</sup> or [Set of objects](#)<sup>c</sup>

**Range**

[Physical object](#)<sup>c</sup>

## has resource location<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/hasResourceLocation>

**Sub Property Of** [location:hasLocation](#)<sup>op</sup>

**Domain**

[Resource](#)<sup>c</sup>

**Range**

[location:Location](#)<sup>c</sup>

## is about<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/isAbout>

**Domain**

[Information](#)<sup>c</sup>

## is realization of<sup>op</sup>

**IRI** <http://w3id.org/CE0N/ontology/resource0DP/isRealizationOf>

**Range**

[Information](#)<sup>c</sup>

## Datatype Properties

|                          |                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------|
| batch size <sup>dp</sup> |                                                                                                                       |
| IRI                      | <a href="http://w3id.org/CE0N/ontology/resource0DP/batchSize">http://w3id.org/CE0N/ontology/resource0DP/batchSize</a> |
| Description              | batchSize intends to represent how many physical objects are belong to a batch of objects.                            |
| Domain                   | <a href="#">Batch of objects</a> <sup>c</sup>                                                                         |
| Range                    | <a href="#">xsd:nonNegativeInteger</a>                                                                                |

## Annotation Properties

|                                          |                                                                                                                       |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| description <sup>ap</sup>                |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/elements/1.1/description">http://purl.org/dc/elements/1.1/description</a>                 |
| contributor <sup>ap</sup>                |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/terms/contributor">http://purl.org/dc/terms/contributor</a>                               |
| created <sup>ap</sup>                    |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/terms/created">http://purl.org/dc/terms/created</a>                                       |
| creator <sup>ap</sup>                    |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/terms/creator">http://purl.org/dc/terms/creator</a>                                       |
| description <sup>ap</sup>                |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/terms/description">http://purl.org/dc/terms/description</a>                               |
| license <sup>ap</sup>                    |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/terms/license">http://purl.org/dc/terms/license</a>                                       |
| title <sup>ap</sup>                      |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/dc/terms/title">http://purl.org/dc/terms/title</a>                                           |
| preferred namespace prefix <sup>ap</sup> |                                                                                                                       |
| IRI                                      | <a href="http://purl.org/vocab/vann/preferredNamespacePrefix">http://purl.org/vocab/vann/preferredNamespacePrefix</a> |

|                                       |                                                                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| preferred namespace uri <sup>ap</sup> |                                                                                                                                                                                               |
| IRI                                   | <a href="http://purl.org/vocab/vann/preferredNamespaceUri">http://purl.org/vocab/vann/preferredNamespaceUri</a>                                                                               |
| covers requirements <sup>ap</sup>     |                                                                                                                                                                                               |
| IRI                                   | <a href="http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements">http://www.ontologydesignpatterns.org/schemas/cannotation-schema.owl#coversRequirements</a> |
| definition <sup>ap</sup>              |                                                                                                                                                                                               |
| IRI                                   | <a href="http://www.w3.org/2004/02/skos/core#definition">http://www.w3.org/2004/02/skos/core#definition</a>                                                                                   |
| pref label <sup>ap</sup>              |                                                                                                                                                                                               |
| IRI                                   | <a href="http://www.w3.org/2004/02/skos/core#prefLabel">http://www.w3.org/2004/02/skos/core#prefLabel</a>                                                                                     |

## Namespaces

|          |                                                                                                                                                         |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| :        | <a href="http://w3id.org/CEON/ontology/resourceODP/">http://w3id.org/CEON/ontology/resourceODP/</a>                                                     |
| dc       | <a href="http://purl.org/dc/elements/1.1/">http://purl.org/dc/elements/1.1/</a>                                                                         |
| dcterms  | <a href="http://purl.org/dc/terms/">http://purl.org/dc/terms/</a>                                                                                       |
| geo      | <a href="http://www.opengis.net/ont/geosparql#">http://www.opengis.net/ont/geosparql#</a>                                                               |
| location | <a href="http://w3id.org/CEON/ontology/location/">http://w3id.org/CEON/ontology/location/</a>                                                           |
| odp      | <a href="http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#">http://www.ontologydesignpatterns.org/schemas/cannotationschema.owl#</a> |
| owl      | <a href="http://www.w3.org/2002/07/owl#">http://www.w3.org/2002/07/owl#</a>                                                                             |
| prov     | <a href="http://www.w3.org/ns/prov#">http://www.w3.org/ns/prov#</a>                                                                                     |
| rdf      | <a href="http://www.w3.org/1999/02/22-rdf-syntax-ns#">http://www.w3.org/1999/02/22-rdf-syntax-ns#</a>                                                   |
| rdfs     | <a href="http://www.w3.org/2000/01/rdf-schema#">http://www.w3.org/2000/01/rdf-schema#</a>                                                               |
| skos     | <a href="http://www.w3.org/2004/02/skos/core#">http://www.w3.org/2004/02/skos/core#</a>                                                                 |
| vann     | <a href="http://purl.org/vocab/vann/">http://purl.org/vocab/vann/</a>                                                                                   |
| xsd      | <a href="http://www.w3.org/2001/XMLSchema#">http://www.w3.org/2001/XMLSchema#</a>                                                                       |

## Legend

|    |                          |
|----|--------------------------|
| c  | Classes                  |
| op | Object<br>Properties     |
| dp | Datatype<br>Properties   |
| ap | Annotation<br>Properties |